

Nonlinear latent manifolds and closure models for projection-based reduced-order models

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May 2, 2026

Abstract

We compare four ANN-enhanced reduced-order modeling routes under aggressive online compression: (i) closure-based intrusive ROMs (PROM-ANN), (ii) latent-manifold intrusive ROMs (PROM-POD-AE), (iii) non-intrusive direct-coefficient surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$, and (iv) non-intrusive latent POD-DL surrogates $(\boldsymbol{\mu}, t) \mapsto \mathbf{z} \mapsto \mathbf{q}_N$. For intrusive models, the online state is obtained by time-discrete least-squares Petrov–Galerkin (LSPG) minimization, solved by Gauss–Newton, and accelerated by ECSW hyperreduction. The closure family includes state-conditioned, parameter–time conditioned, and hybrid maps for truncated coefficients. The latent family evolves low-dimensional latent coordinates and reconstructs reduced coefficients through a decoder.

A central methodological choice is ROM-consistent training: learning targets are generated by a linear n_{tot} -dimensional PROM/HPROM solve in the full truncated space, rather than by direct projection of HDM trajectories. We also consider a second enrichment stage built from additional low-cost HPROM trajectories at Latin-hypercube parameter points. This manuscript consolidates the mathematical formulation and experimental protocol for the 250×250 2D parametric inviscid Burgers benchmark and prepares an extension to a shallow-water benchmark.

1 Introduction

Projection-based model order reduction (PMOR) is a standard framework for reducing high-dimensional parametric models while preserving governing equations [1, 2]. For nonlinear transport-dominated dynamics, linear subspaces may require large dimensions, which motivates nonlinear manifolds, closure modeling, and latent representations [3, 4, 5].

From a machine-learning viewpoint, four families are relevant to this work:

- intrusive closure-enhanced projection (PROM-ANN) [3],
- intrusive latent-manifold projection (PROM-POD-AE), in the spirit of hyperreduced nonlinear manifold ROMs [6],
- non-intrusive direct-coefficient regression maps $(\boldsymbol{\mu}, t) \mapsto \mathbf{q}_N$,
- non-intrusive latent POD-DL-ROM maps [7].

Naming convention used in this manuscript. When we say **POD-DL**, we refer to the non-intrusive family in the sense of Fresca–Manzoni [7]. When we say **PROM-POD-AE**, we refer to an intrusive latent-manifold projection route (conceptually aligned with intrusive nonlinear-manifold HPROM literature such as [6]), even though those references use different method names.

The goal is a backend-consistent comparison under strong online compression, using the same HDM residual, same time discretization, and same intrusive solver/hyperreduction stack for intrusive variants.

2 Linear PROM/HPROM Reference and Notation

2.1 POD basis from SVD

Let $\mathbf{u}(t; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM state with parameters $\boldsymbol{\mu} \in \mathcal{D} \subset \mathbb{R}^p$. Define

$$\mathbf{u}_{\text{ref}} := \frac{1}{N_s} \sum_{i=1}^{N_s} \mathbf{u}^i, \quad \mathbf{S}_c = [\mathbf{u}^1 - \mathbf{u}_{\text{ref}} \cdots \mathbf{u}^{N_s} - \mathbf{u}_{\text{ref}}]. \quad (1)$$

Let

$$\mathbf{S}_c = \mathbf{U}_S \boldsymbol{\Sigma}_S \mathbf{Y}_S^T \quad (2)$$

be its thin SVD, and define the linear ROB

$$\mathbf{V}_{\text{tot}} \in \mathbb{R}^{N \times n_{\text{tot}}} \quad (3)$$

as the first n_{tot} left singular vectors of $\mathbf{U}_S$.

The linear reduced reconstruction is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N(t; \boldsymbol{\mu}), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \in \mathbb{R}^{n_{\text{tot}}}. \quad (4)$$

2.2 Linear time-discrete LSPG (PROM)

Let $\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) \in \mathbb{R}^N$ denote the HDM time-discrete residual at step $m+1$. The linear PROM solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (5)$$

With

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_N := \mathbf{V}_{\text{tot}}, \quad \mathbf{W}_N^{m+1,k} := \mathbf{J}^{m+1,k} \mathbf{T}_N, \quad (6)$$

the Gauss–Newton increment solves

$$(\mathbf{W}_N^{m+1,k})^T \mathbf{W}_N^{m+1,k} \Delta \mathbf{q}_N^{m+1,k} = -(\mathbf{W}_N^{m+1,k})^T \mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{q}_N^{m+1,k}; \boldsymbol{\mu}). \quad (7)$$

2.3 Linear HPROM via ECSW

Assume element-wise residual decomposition

$$\mathbf{r}^{m+1}(\mathbf{u}; \boldsymbol{\mu}) = \sum_{e_i \in \mathcal{E}} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i} \mathbf{u}; \boldsymbol{\mu}). \quad (8)$$

ECSW approximation yields

$$\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}) = \sum_{e_i \in \tilde{\mathcal{E}}} \xi_{e_i} \mathbf{L}_{e_i}^T \mathbf{r}_{e_i}^{m+1}(\mathbf{L}_{e_i} [\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathbf{x}]; \boldsymbol{\mu}), \quad \xi_{e_i} > 0, \quad (9)$$

with sampled set $\tilde{\mathcal{E}} \subset \mathcal{E}$ [8, 9]. The linear HPROM solve is

$$\mathbf{q}_N^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^{n_{\text{tot}}}} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2. \quad (10)$$

2.4 ROM-consistent training targets

All learning targets in this work are extracted from trajectories $\mathbf{q}_N(t; \boldsymbol{\mu})$ generated by the linear ROM solves (5) or (10), depending on the selected backend.

3 Method I: PROM-ANN Closure Models

3.1 Manifold definitions

For closure models, we introduce the partition

$$\mathbf{V}_{\text{tot}} = [\mathbf{V} \ \overline{\mathbf{V}}], \quad \mathbf{q}_N = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}, \quad n + \bar{n} = n_{\text{tot}}. \quad (11)$$

PROM-ANN uses retained coordinates $\mathbf{q} \in \mathbb{R}^n$ as online unknowns and reconstructs truncated coordinates via learned maps:

$$\text{Case 1: } \bar{\mathbf{q}} = \mathcal{N}(\mathbf{q}), \quad (12)$$

$$\text{Case 2: } \bar{\mathbf{q}} = \mathcal{M}(\boldsymbol{\mu}, t), \quad (13)$$

$$\text{Case 3: } \bar{\mathbf{q}} = \mathcal{H}(\mathbf{q}, \boldsymbol{\mu}, t). \quad (14)$$

The state approximation is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V} \mathbf{q}(t; \boldsymbol{\mu}) + \overline{\mathbf{V}} \mathcal{F}(\mathbf{q}, \boldsymbol{\mu}, t), \quad (15)$$

with $\mathcal{F} \in \{\mathcal{N}, \mathcal{M}, \mathcal{H}\}$.

3.2 LSPG/Gauss–Newton solve

At each time step,

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{x}; \boldsymbol{\mu}); \boldsymbol{\mu})\|_2^2. \quad (16)$$

Define

$$\mathbf{J}^{m+1,k} := \frac{\partial \mathbf{r}^{m+1}}{\partial \mathbf{u}}, \quad \mathbf{T}_q^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{q}} = \mathbf{V} + \overline{\mathbf{V}} \frac{\partial \mathcal{F}}{\partial \mathbf{q}}, \quad \mathbf{W}_q^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_q^{m+1,k}. \quad (17)$$

The Gauss–Newton increment solves

$$(\mathbf{W}_q^{m+1,k})^T \mathbf{W}_q^{m+1,k} \Delta \mathbf{q}^{m+1,k} = -(\mathbf{W}_q^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{q}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (18)$$

3.3 ECSW hyperreduced form

PROM-ANN HPRM solves

$$\mathbf{q}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{x} \in \mathbb{R}^n} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{x}; \boldsymbol{\mu})\|_2^2, \quad (19)$$

with test basis $\widetilde{\mathbf{W}}_q = \widetilde{\mathbf{J}} \mathbf{T}_q$.

3.4 Offline training target

For Cases 1–3, the supervised targets are extracted from ROM-consistent $\mathbf{q}_N = [\mathbf{q}; \bar{\mathbf{q}}]$ trajectories:

- Case 1: train $\mathcal{N}$ from $(\mathbf{q}, \bar{\mathbf{q}})$,
- Case 2: train $\mathcal{M}$ from $(\boldsymbol{\mu}, t, \bar{\mathbf{q}})$,
- Case 3: train $\mathcal{H}$ from $(\mathbf{q}, \boldsymbol{\mu}, t, \bar{\mathbf{q}})$.

4 Method II: PROM-POD-AE Latent Manifold

4.1 Autoencoder and latent manifold

Let

$$\mathbf{z} = \mathcal{E}(\mathbf{q}_N), \quad \hat{\mathbf{q}}_N = \mathcal{D}(\mathbf{z}), \quad \mathbf{z} \in \mathbb{R}^{n_z}, \quad n_z \ll n_{\text{tot}}. \quad (20)$$

The intrusive latent manifold is

$$\tilde{\mathbf{u}}(t; \boldsymbol{\mu}) = \mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{z}(t; \boldsymbol{\mu})). \quad (21)$$

This is an intrusive latent-manifold projection approach, in line with intrusive nonlinear-manifold HPRM literature [6], and is separated from the non-intrusive POD-DL-ROM line [7].

4.2 LSPG/Gauss–Newton solve in latent coordinates

The online unknown is $\mathbf{z}^{m+1} \in \mathbb{R}^{n_z}$:

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \|\mathbf{r}^{m+1}(\mathbf{u}_{\text{ref}} + \mathbf{V}_{\text{tot}} \mathcal{D}(\mathbf{y}); \boldsymbol{\mu})\|_2^2. \quad (22)$$

Define

$$\mathbf{T}_z^{m+1,k} := \frac{\partial \tilde{\mathbf{u}}^{m+1}}{\partial \mathbf{z}} = \mathbf{V}_{\text{tot}} \frac{\partial \mathcal{D}}{\partial \mathbf{z}}, \quad \mathbf{W}_z^{m+1,k} = \mathbf{J}^{m+1,k} \mathbf{T}_z^{m+1,k}. \quad (23)$$

The Gauss–Newton step solves

$$(\mathbf{W}_z^{m+1,k})^T \mathbf{W}_z^{m+1,k} \Delta \mathbf{z}^{m+1,k} = -(\mathbf{W}_z^{m+1,k})^T \mathbf{r}^{m+1}(\tilde{\mathbf{u}}^{m+1}(\mathbf{z}^{m+1,k}; \boldsymbol{\mu}); \boldsymbol{\mu}). \quad (24)$$

4.3 ECSW hyperreduced form

PROM-POD-AE HPRM solves

$$\mathbf{z}^{m+1}(\boldsymbol{\mu}) = \arg \min_{\mathbf{y} \in \mathbb{R}^{n_z}} \|\tilde{\mathbf{r}}^{m+1}(\mathbf{y}; \boldsymbol{\mu})\|_2^2, \quad (25)$$

with test basis $\tilde{\mathbf{W}}_z = \tilde{\mathbf{J}} \mathbf{T}_z$.

5 Method III: Data-Driven Direct-Coefficient Baseline

5.1 Model definition

The non-intrusive baseline maps exogenous inputs directly to reduced coefficients:

$$\mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{G}_q(\boldsymbol{\mu}, t). \quad (26)$$

This model does not require online residual minimization.

5.2 Training objective and online cost

Given ROM-consistent targets $\mathbf{q}_N^{\text{ref}}(t; \boldsymbol{\mu})$, one minimizes a supervised loss (e.g., MSE)

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{samp}}} \sum_{j=1}^{N_{\text{samp}}} \left\| \mathcal{G}_q(\boldsymbol{\mu}_j, t_j) - \mathbf{q}_{N,j}^{\text{ref}} \right\|_2^2. \quad (27)$$

Online prediction is then a direct forward evaluation of $\mathcal{G}_q$ followed by reconstruction with $\mathbf{V}_{\text{tot}}$.

6 Method IV: Data-Driven Latent POD-DL Baseline

6.1 Model definition

The second non-intrusive baseline follows the POD-DL family [7]:

$$\mathbf{z}(t; \boldsymbol{\mu}) \approx \mathcal{G}_z(\boldsymbol{\mu}, t), \quad \mathbf{q}_N(t; \boldsymbol{\mu}) \approx \mathcal{D}(\mathcal{G}_z(\boldsymbol{\mu}, t)), \quad (28)$$

where $\mathcal{D}$ is a decoder trained offline and $\mathbf{z} \in \mathbb{R}^{n_z}$.

6.2 Training objective and online cost

This baseline is trained non-intrusively through supervised latent/dynamic losses (with reconstruction consistency through $\mathcal{D}$), and evaluated online by a single forward pass of $\mathcal{G}_z$ and $\mathcal{D}$, without residual minimization.

7 Optional Physics-Regularized Training Note

Physics-regularized neural training adds a residual-based term to a supervised objective [10, 11]. In this project, such variants are treated as optional diagnostics (not core baselines) because current results are mixed and significantly more expensive offline.

8 Burgers Benchmark and Experimental Protocol

8.1 Governing equations and parameter domain

We use the 2D parametric inviscid Burgers benchmark from [3], in conservative vector form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = \mathbf{S}(x; \boldsymbol{\mu}), \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad (29)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2}u_x^2 \\ \frac{1}{2}u_x u_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} \frac{1}{2}u_x u_y \\ \frac{1}{2}u_y^2 \end{bmatrix}, \quad \mathbf{S}(x; \boldsymbol{\mu}) = \begin{bmatrix} 0.02 \exp(\mu_2 x) \\ 0 \end{bmatrix}. \quad (30)$$

The problem is defined on $\Omega = [0, 100] \times [0, 100]$ with final time $T_f = 25$ and parameters $\boldsymbol{\mu} = (\mu_1, \mu_2) \in \mathcal{D}$, where

$$\mathcal{D} = [4.25, 5.50] \times [0.015, 0.03]. \quad (31)$$

This conservative form is the one used by the discrete residual and Jacobian in the ROM/HPROM pipeline. The initial/boundary conditions used in our campaign are

$$u_x(0, y, t; \boldsymbol{\mu}) = \mu_1, \quad u_x(x, y, 0) = 1, \quad u_y(x, y, 0) = 1. \quad (32)$$

8.2 Numerical discretization

The HDM uses a cell-centered finite-volume Godunov discretization on a 250×250 mesh, hence $N = 2N_x N_y = 125,000$. Time integration uses 500 uniform steps with $\Delta t = 0.05$ (thus 501 stored states including $t = 0$). The same time-discrete residual is used by linear PROM/HPROM and by all intrusive ANN-enhanced variants (PROM-ANN and PROM-POD-AE) through LSPG/Gauss–Newton solves.

8.3 Training and evaluation protocol

Stage 1 builds $\mathbf{V}_{\text{tot}}$ from HDM snapshots at 9 structured training parameters (a 3×3 tensor grid on $\mathcal{D}$), giving $N_s = 9 \times 501 = 4509$ snapshots. Stage 2 generates ROM-consistent coefficients $\mathbf{q}_N(t; \boldsymbol{\mu})$ by solving the linear full-space reduced model (n_{tot}). Stage 3 trains the learned maps from these reduced targets.

Evaluation is reported at one verification point and two off-grid points:

$$\boldsymbol{\mu}^{(v)} = (4.875, 0.0225), \quad \boldsymbol{\mu}^{(1)} = (4.56, 0.019), \quad \boldsymbol{\mu}^{(2)} = (5.19, 0.026). \quad (33)$$

An enrichment stage adds 20 Latin-hypercube points in parameter space. The enrichment data is generated with reduced solves (no additional HDM runs), then the same model families are retrained and re-evaluated at the same three points.

8.4 Backend-consistency policy

For this manuscript branch, intrusive comparisons are HPROM-consistent end-to-end:

- Stage-2 reduced targets generated with linear HPROM in $\mathbf{V}_{\text{tot}}$,
- Stage-3 map training from HPROM-consistent datasets,
- online intrusive evaluation with HPROM backend.

8.5 Reported quantities

- trajectory relative errors vs HDM,
- coefficient-space errors vs linear n_{tot} reference,
- per-index absolute/relative coefficient curves,
- time-resolved coefficient heatmaps,
- online runtimes and speedups.

9 Shallow-Water Benchmark and Experimental Protocol

9.1 Governing equations and parameterized initial condition

The second benchmark is a 2D shallow-water model on a flat bottom, in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad \mathbf{U} = \begin{bmatrix} h \\ hu_x \\ hu_y \end{bmatrix}, \quad (34)$$

with

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} hu_x \\ hu_x^2 + \frac{1}{2}gh^2 \\ hu_xu_y \end{bmatrix}, \quad \mathbf{G}(\mathbf{U}) = \begin{bmatrix} hu_y \\ hu_xu_y \\ hu_y^2 + \frac{1}{2}gh^2 \end{bmatrix}. \quad (35)$$

The parametric scenario is a collision of two Gaussian bumps in water depth:

$$h(x, y, 0; \boldsymbol{\mu}) = h_0 + \mu_1 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_1\|_2^2}{2\sigma^2}\right) + \mu_2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{c}_2\|_2^2}{2\sigma^2}\right), \quad (36)$$

with zero initial velocity components ($u_x = u_y = 0$), $h_0 = 1$, $\sigma = 0.045$, $\mathbf{c}_1 = (0.30, 0.50)$, $\mathbf{c}_2 = (0.70, 0.50)$, and

$$\boldsymbol{\mu} = (\mu_1, \mu_2) \in [0.06, 0.14] \times [0.06, 0.14]. \quad (37)$$

Reflective boundary conditions are imposed on all sides.

9.2 Numerical setup

The reference configuration in `shallowwaters-rom-workbench` uses $N_x = N_y = 600$ on $\Omega = [0, 1] \times [0, 1]$, so $N = 3N_x N_y$ unknowns. The HDM uses MUSCL reconstruction with HLL/HLLC interface fluxes and an implicit BDF2 time integrator (BE startup). The default fixed time step is $\Delta t = 7 \times 10^{-5}$ up to $T_f = 0.28$, yielding 4001 stored states per parameter value.

9.3 Sampling protocol aligned with Burgers

The staged workflow mirrors Burgers:

- baseline set: 9 structured parameter points (3×3 grid),
- enrichment set: 20 LHS points,
- Stage 1: POD basis $\mathbf{V}_{\text{tot}}$ from HDM snapshots,
- Stage 2: ROM-consistent $\mathbf{q}_N$ dataset from linear HPROM in $\mathbf{V}_{\text{tot}}$,
- Stage 3: train PROM-ANN (Cases 1–3), PROM-POD-AE, and both data-driven baselines (direct-coefficient and latent POD-DL),
- online intrusive evaluation with HPROM backend.

This keeps the methodology consistent across both benchmarks so that differences in performance are attributable to the physics and manifold structure, not to pipeline changes.

10 Scope for the Current Draft

- Core model families: linear HPROM reference, PROM-ANN (Cases 1–3), PROM-POD-AE, data-driven direct-coefficient, and data-driven latent POD-DL-ROM.
- Baseline/enriched comparison is retained as a controlled data axis.
- PG/PRNN-style variants are currently kept as optional diagnostic tracks.

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